This document replaces Section 2.1 in the SDG Methodology document

PHASE 1: Defining the scope and application for the case study.

The first phase of the SDG screening assessment aims to clarify the case study's scope and application. Before you can dig into the case, i.e. understanding the details and finding data, you will have to discuss some aspects of the case.

The case provides a description of a shift from one situation (the baseline system) to another (the new system), defined by some technological differences. The case is too wide to embrace as it is and, therefore, not feasible for carrying out an assessment. So, you must select a subset of what is covered in the case description for your study, limiting the scope on several parameters. The following aspects should be considered for scoping the study:

- **Time span**. Are you looking at an old technology that has been phased out compared to present technology? Then, you may select points in time where the old one was dominating and where the present is dominating, thereby establishing a clear-cut instead of working with systems of mixed technologies. Remember, the societal framework also changes over time. E.g. 20 years ago in Denmark, 1 kWh of power was produced mainly from burning fossil coal, while coal is almost absent today.
    - Another approach to time span is to look at two future scenarios representing the two technologies at the same point in time, e.g. 2030. Here, it will be crucial to establish comparable frameworks for the competing technologies to compile a fair assessment.
- **Application span**. How big a part of the system do you want to cover? If you do not limit yourself to some sub-system level, you will drown in data and find it difficult to reach conclusions. If you limit yourself too narrowly, you may cover only a tiny part of the core technology shift.
- **Geographical span**. Cases may cover technologies that are globally relevant. However, you cannot assess the entire world. Therefore, selecting a specific country, region or even a very local area will make the study feasible regarding data collection and understanding the specific physical boundaries of the systems.

An example is the shift from using fossil-based chemicals to using bio-based chemicals. You would not be able to assess all chemicals that may be produced from either fossil or bio sources in one assessment. Also, you could never cover all the applications of a group of chemicals nor look at their use over the last century. So, you will have to discuss in the group and conclude how you would like to limit the scope of your study to make it feasible to carry out.

Furthermore, to judge positive or negative impacts on sustainable development, one must consider the expected outcomes of the technology shift and translate it into a concrete application in socio-economic systems.

A technology shift may lead to a replacement of existing technologies, but it may also lead to the introduction of technologies that did not exist before, and some of these may even create new needs in society and new demands in the market.

During the study, we want to focus the assessment on the applications resulting from the project outcomes that are most likely to enter the market and have the largest potential impact on the SDGs (society or environment).

Eventually, our ability to predict which applications will take place and to what extent they will be effective is limited. Therefore, our choice of considered applications may be accompanied by large uncertainties or even be regarded as arbitrary. It is, thus, important to document them transparently, arguing for the different assumptions made.

To sum up, the output from Phase 1 is a brief report describing the scope limitations you have selected for your study regarding time, application, and geography. This report forms the dynamic basis for your further investigations into the case in phases 2-5. As the assessment methodology is highly iterative, the basis is to be seen as the first iteration of how you approach the full study.

/ Adjusted by M. Sci. Eng. Christian Poll, DTU Sustain, June 2024 /